

The Rosei Interband ϵ_2 of Gold, Rederived in Rosei's Own Notation

A self-contained construction from Fermi's Golden Rule in k-space that recreates Guerrisi–Rosei–Winsemius Eqs. (1)–(10), with corrected (4'), (6'), (8')

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Abstract

Starting from Fermi's Golden Rule (FGR) for Bloch electrons and the linearization $(u, v) = (k_\perp^2, k_\parallel^2)$ of my notes, the interband $\epsilon_2(\hbar\omega, T)$ of gold is rebuilt *in the exact symbols of Guerrisi, Rosei and Winsemius* [Phys. Rev. B **12**, 557 (1975), “GRW”]: the band pairs of their Eqs. (1)–(2), the constant-energy-difference surface (CEDS) of Eq. (3), the energy-distributed joint density of states (EDJDOS) $D_{l \rightarrow u}(E, \hbar\omega)$ of Eq. (4), the mass parameter $\mathcal{F}_{l \rightarrow u}$ of Eq. (5), the surface parameterization $k_\parallel(E, \hbar\omega)$ of Eq. (6), the thermally weighted integral $\mathcal{I}(\hbar\omega, T)$ of Eq. (7), the limit $E_{\max}$ of Eq. (8), the assembled ϵ_2 of Eq. (9), and the strengths $S = \mathcal{F}|P|^2$ of Eq. (10). Three printed equations do not survive the reconstruction and are corrected here: (6) is dimensionally garbled, (8) has its $u \parallel l$ subscripts interchanged, and (4) — this is a new finding — is inconsistent with (9) for *any* standard matrix-element convention; the FGR-exact replacement is $D_{l \rightarrow u} = (16\pi^2 \mathcal{D}_{l \rightarrow u} k_\parallel)^{-1} = \mathcal{F}^2 / (4\pi^2 \hbar^4 k_\parallel)$, a discrepancy that is constant per critical point and therefore invisibly absorbed by the fitted strengths. The L point, which GRW delegate to earlier papers, is derived with the same machinery. The document closes with the photonic density of states and the spontaneous-emission spectrum in the same notation, verified by the exact closure $R(\hbar\omega) = (\omega^3 / \pi^2 \hbar c^3) \epsilon_2 n_B$. Gaussian units throughout, as in the original paper; E is measured from the Fermi level.

Contents

1	Notation (Rosei's, fixed once)	2
2	Fermi's Golden Rule in k-space	2
3	The X point	3
3.1	Bands and CEDS: GRW (1)–(3)	3
3.2	Linearization and the Jacobian	3
3.3	The EDJDOS: GRW (4)–(5), with (4) corrected	3
3.4	Topology, integration limits, and GRW (7)–(8)	4
4	The L point (delegated by GRW; same machinery)	5
5	Assembly: GRW (9)–(10)	6
6	Photonic density of states and the emission spectrum	7
6.1	Mode counting	7
6.2	Spontaneous emission of one pair	7
6.3	The interband PL spectrum and the closure check	7

1 Notation (Rosei's, fixed once)

E	final-state (upper-band) energy, from $E_F = 0$; $\hbar\omega$: photon energy
$\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+}$	level distances from E_F at X (X_6^- above, X_7^+ below); likewise $\hbar\omega_{L_6^-}, \hbar\omega_{L_7^+}$
$\mathcal{E}_g^X, \mathcal{E}_g^L$	direct gaps, $\mathcal{E}_g^X = \hbar\omega_{X_6^-} + \hbar\omega_{X_7^+}$, $\mathcal{E}_g^L = \hbar\omega_{L_6^-} + \hbar\omega_{L_7^+}$
$m_{u\perp}, m_{u\parallel}, m_{l\perp}, m_{l\parallel}$	effective masses (positive; signs written explicitly)
A_i, B_i	$\hbar^2/2m_{i\perp}, \hbar^2/2m_{i\parallel}$ ($i = u, l$): curvature shorthands for derivations
$\Omega_{lu}(\mathbf{k})$	transition energy $\hbar\omega_u - \hbar\omega_l$
$D_{l\rightarrow u}, \mathcal{F}_{l\rightarrow u}, \mathcal{D}_{l\rightarrow u}$	EDJDOS, its mass parameter, and the Jacobian determinant
$\mathcal{I}(\hbar\omega, T)$	thermally weighted EDJDOS integral, GRW Eq. (7)
$ P ^2$	$ \langle u \hat{\mathbf{e}}\cdot\nabla l\rangle ^2$ averaged (length ⁻² ; Sec. 2)
$N_X = 6, N_L = 8$	equivalent half-neighborhoods (GRW's counting)
$f(E, T), n_B(\hbar\omega)$	Fermi–Dirac and Bose–Einstein functions

Gaussian units (e^2 has dimension energy $\times$ length), matching the 1975 paper.

2 Fermi's Golden Rule in k-space

Minimal coupling in the Coulomb gauge gives $\hat{H}_{\text{int}} = (e/mc) \mathbf{A}\cdot\hat{\mathbf{p}}$ (the A^2 term is interband-inert). For a classical wave $\mathbf{E} = \hat{\mathbf{e}}E_0 \cos \omega t$, i.e. $\mathbf{A} = -(cE_0/\omega)\hat{\mathbf{e}} \sin \omega t$, the absorption half of $\hat{H}_{\text{int}}$ has amplitude $(eE_0/2m\omega) \hat{\mathbf{e}}\cdot\hat{\mathbf{p}}$, and photon momentum is negligible on the zone scale, so transitions are vertical. FGR per initial Bloch state, summed over $\mathbf{k}$ (spin explicit), balanced against the Gaussian dissipation rate $\langle \dot{U} \rangle = \omega \epsilon_2 E_0^2 / 8\pi$ per unit volume, yields the exact starting point

$$\epsilon_2(\omega) = \frac{4\pi^2 e^2}{m^2 \omega^2} \frac{2}{(2\pi)^3} \int_{\text{BZ}} d^3k |\hat{\mathbf{e}}\cdot\mathbf{p}_{ul}(\mathbf{k})|^2 \delta(\hbar\omega_u(\mathbf{k}) - \hbar\omega_l(\mathbf{k}) - \hbar\omega) [f(\hbar\omega_l) - f(\hbar\omega_u)]. \quad (2.1)$$

Near a critical point Rosei freezes the matrix element and averages over polarization. Writing it, as in the older literature, as a matrix element of the gradient, $\mathbf{p}_{ul} = -i\hbar\langle u|\nabla|l\rangle$, and defining

$$|P|^2 \equiv |\langle u|\nabla|l\rangle|^2 \quad (\text{dimension length}^{-2}), \quad \overline{|\hat{\mathbf{e}}\cdot\mathbf{p}_{ul}|^2} = \frac{\hbar^2 |P|^2}{3}, \quad (2.2)$$

equation (2.1) becomes, per critical-point family,

$$\epsilon_2(\hbar\omega, T) = \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot \frac{2}{(2\pi)^3} N |P|^2 \int_{\text{half-nbhd}} d^3k \delta(\Omega_{lu}(\mathbf{k}) - \hbar\omega) [f(E - \hbar\omega) - f(E)] \Big|_{E=\hbar\omega_u(\mathbf{k})}, \quad (2.3)$$

where N counts equivalent *half*-neighborhoods ($N_X = 6, N_L = 8$: each zone face carries half of a full CEDS; this convention must be paired with the half-space EDJDOS below, on pain of a silent factor 2). The occupation factor is the full $f(E - \hbar\omega) - f(E)$; for d bands sunk $\gtrsim 1.5$ eV below E_F it reduces to GRW's $[1 - f(E, T)]$, and the general form is kept for the non-equilibrium extensions (f^S, f^P).

3 The X point

3.1 Bands and CEDS: GRW (1)–(3)

GRW’s parabolic pair at X , in their exact form (energies from E_F ; $k_{\parallel}$ along $\Delta = \Gamma X$, $k_{\perp}$ in the zone face):

$$E = \hbar\omega_u(\mathbf{k}) = \hbar\omega_{X_6^-} + \frac{\hbar^2 k_{\perp}^2}{2m_{u\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{u\parallel}}, \quad [\text{GRW (1)}] \quad (3.1)$$

$$E - \hbar\omega = \hbar\omega_l(\mathbf{k}) = -\hbar\omega_{X_7^+} - \frac{\hbar^2 k_{\perp}^2}{2m_{l\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{l\parallel}}, \quad [\text{GRW (2)}] \quad (3.2)$$

i.e. the upper X_6^- band is a saddle (rises in the face, falls along Δ) and the lower X_7^+ band is a maximum. Vertical transitions of energy $\hbar\omega$ live on the constant-energy-difference surface

$$\Omega_{lu}(\mathbf{k}) - \hbar\omega = \hbar\omega_u(\mathbf{k}) - \hbar\omega_l(\mathbf{k}) - \hbar\omega = 0. \quad [\text{GRW (3)}] \quad (3.3)$$

Subtracting (3.2) from (3.1), with $A_i = \hbar^2/2m_{i\perp}$, $B_i = \hbar^2/2m_{i\parallel}$,

$$\Omega_{lu} = \mathcal{E}_g^X + \bar{A}_X k_{\perp}^2 + \bar{B}_X k_{\parallel}^2, \quad \bar{A}_X = A_u + A_l > 0, \quad \bar{B}_X = B_l - B_u. \quad (3.4)$$

GRW’s own apparatus (open CEDS, sub-gap tail, the $-20k_B T$ floor of their Eq. (7), the step-like ~ 1.8 eV onset) requires $\boxed{\bar{B}_X < 0}$, i.e. $m_{u\parallel} < m_{l\parallel}$: the sp band falls along Δ *faster* than the d band — the flat- d prior. The main text follows GRW’s branch; see Remark 3.2 for the status of my own C&S extraction, which currently has the opposite $\parallel$ assignment.

3.2 Linearization and the Jacobian

With $u = k_{\perp}^2$, $v = k_{\parallel}^2$ ($u, v \geq 0$), the two constraints — final-state energy E and transition energy $\hbar\omega$ — are linear:

$$\begin{pmatrix} E - \hbar\omega_{X_6^-} \\ \hbar\omega - \mathcal{E}_g^X \end{pmatrix} = \begin{pmatrix} A_u & -B_u \\ \bar{A}_X & \bar{B}_X \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}, \quad \det = A_u \bar{B}_X + B_u \bar{A}_X = A_u B_l + A_l B_u \equiv \mathcal{D}_X > 0 \quad (3.5)$$

— note that $\mathcal{D}_X$ is positive for *either* sign of $\bar{B}_X$. Inverting,

$$u = \frac{\bar{B}_X (E - \hbar\omega_{X_6^-}) + B_u (\hbar\omega - \mathcal{E}_g^X)}{\mathcal{D}_X}, \quad v = k_{\parallel}^2 = \frac{A_u (\hbar\omega - \mathcal{E}_g^X) - \bar{A}_X (E - \hbar\omega_{X_6^-})}{\mathcal{D}_X}. \quad (3.6)$$

The second line, restored to GRW’s mass symbols via $\mathcal{D}_X = (\hbar^2/2)^2 \mathcal{F}^{-2}$ (Eq. (3.11) below), is their Eq. (6):

$$k_{\parallel}^2 = \frac{2\mathcal{F}^2}{\hbar^2} \left[\frac{\hbar\omega - \hbar\omega_{X_7^+} - E}{m_{u\perp}} + \frac{\hbar\omega_{X_6^-} - E}{m_{l\perp}} \right]. \quad [\text{GRW (6) corrected}] \quad (3.7)$$

As printed, (6) lacks the $1/m_{u\perp}$ on its first term, the overall $2\mathcal{F}^2/\hbar^2$, and the square on $k_{\parallel}$ (it equates a wavevector to an energy); the three numerator slots of (3.7) match the print term by term, so (3.7) is the intended equation.

3.3 The EDJDOS: GRW (4)–(5), with (4) corrected

Rosei’s central object is the number of ($l \rightarrow u$) transition pairs per unit crystal volume, per unit final energy E , per unit transition energy $\hbar\omega$, for one half-neighborhood and one spin:

$$D_{l \rightarrow u}(E, \hbar\omega) \equiv \frac{1}{(2\pi)^3} \int_{k_{\parallel} > 0} d^3k \, \delta(\hbar\omega_u(\mathbf{k}) - E) \delta(\Omega_{lu}(\mathbf{k}) - \hbar\omega). \quad (3.8)$$

Axial symmetry gives $d^3k = 2\pi k_\perp dk_\perp dk_\parallel$; with $du = 2k_\perp dk_\perp$, $dv = 2k_\parallel dk_\parallel$,

$$\int_{k_\parallel > 0} d^3k = \frac{\pi}{2} \int_0^\infty du \int_0^\infty \frac{dv}{\sqrt{v}}, \quad (3.9)$$

and the linear map (3.5) collapses both δ 's with Jacobian $1/\mathcal{D}_X$:

$$\boxed{D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 \mathcal{D}_{l \rightarrow u}} \frac{1}{k_\parallel(E, \hbar\omega)} = \frac{\mathcal{F}_{l \rightarrow u}^2}{4\pi^2 \hbar^4} \frac{1}{k_\parallel(E, \hbar\omega)}} \quad [\text{GRW (4) corrected}] \quad (3.10)$$

with $k_\parallel(E, \hbar\omega)$ from (3.7), supported on $u, v \geq 0$, and

$$\mathcal{F}_{l \rightarrow u} = \frac{\hbar^2}{2\sqrt{\mathcal{D}_{l \rightarrow u}}} = \left[\frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2}. \quad [\text{GRW (5)}] \quad (3.11)$$

Remark 3.1 (Correction of the printed (4)) *GRW print $D_{l \rightarrow u} = (8\pi^2 \hbar^2)^{-1} \mathcal{F}/k_\parallel$. No standard convention rescues it: with $|P|^2$ a squared ∇ matrix element (or a squared momentum), the pair {printed (4), printed (9)} is dimensionally inconsistent, while (3.10) — which is forced by the definition (3.8) — closes the FGR chain exactly (Sec. 5). The two forms differ by the factor $2\mathcal{F}/\hbar^2$, a constant for each critical point: it cannot affect any line shape and is silently absorbed into the fitted strength $S = \mathcal{F}|P|^2$ of GRW's Eq. (10), which is why the defect has no observable consequence in their (or my) fits. The first equality of (3.10) already appears in my note Interband Absorption at X and L; the present derivation shows it is the only correct one.*

3.4 Topology, integration limits, and GRW (7)–(8)

On GRW's branch $\bar{B}_X < 0$ the CEDS is an open hyperboloid for every $\hbar\omega$: transitions exist on both sides of $\mathcal{E}_g^X$, and along the CEDS E decreases monotonically away from its top edge ($dE \propto -\mathcal{D}_X dv < 0$), unboundedly. The window is therefore cut from below by occupation alone, for which GRW adopt the practical floor $E_{\min} = -20k_B T$, and from above by the geometric edge $E_{\max}(\hbar\omega)$, which changes character at the gap:

Above the gap ($\hbar\omega \geq \mathcal{E}_g^X$): the edge is the $k_\parallel \rightarrow 0$ circle ($v = 0$ in (3.6)), where $D_{l \rightarrow u}$ of (3.10) diverges as an inverse square root:

$$E_{\max}^> = \hbar\omega_{X_6^-} + \frac{A_u}{\bar{A}_X} (\hbar\omega - \mathcal{E}_g^X) = \hbar\omega_{X_6^-} + \frac{m_{l\perp}}{m_{u\perp} + m_{l\perp}} (\hbar\omega - \mathcal{E}_g^X), \quad k_\parallel^2 = \frac{\bar{A}_X}{\mathcal{D}_X} (E_{\max}^> - E). \quad (3.12)$$

Below the gap ($\hbar\omega < \mathcal{E}_g^X$): the $k_\parallel = 0$ circle no longer exists ($v > 0$ everywhere on the CEDS); the edge is the $k_\perp = 0$ point ($u = 0$ in (3.6)), where $k_\parallel$ stays finite and $D_{l \rightarrow u}$ is finite:

$$E_{\max}^< = \hbar\omega_{X_6^-} - \frac{B_u}{|\bar{B}_X|} (\mathcal{E}_g^X - \hbar\omega) = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega) \frac{m_{l\parallel}}{m_{u\parallel} - m_{l\parallel}}. \quad [\text{GRW (8) corrected}] \quad (3.13)$$

The printed (8) carries $m_{u\parallel}/(m_{l\parallel} - m_{u\parallel})$: the $u \parallel / l \parallel$ subscripts are interchanged (one consistent swap; the subscripts of (6) are *not* swapped, so the defect is specific to (8)). As printed, (8) would place the sub-gap edge *above* $\hbar\omega_{X_6^-}$, where the sub-gap CEDS has no states.

The finite edge (3.13) is what makes the X onset *step-like* (GRW's wording) rather than divergent: at $T = 0$ absorption begins when $E_{\max}^<$ crosses E_F ,

$$\hbar\omega_{\text{on}} = \mathcal{E}_g^X - \hbar\omega_{X_6^-} \frac{m_{l\parallel} - m_{u\parallel}}{m_{l\parallel}} \simeq 1.94 - 0.17 \times \frac{0.40 - 0.15}{0.40} = 1.83 \text{ eV}, \quad (3.14)$$

the piezomodulation threshold, with the famous low-energy tail filling 1.83–1.94 eV geometrically (not by broadening).

Finally the thermally weighted integral, exactly as GRW write it:

$$\mathcal{I}(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE, \quad [\text{GRW (7)}] \quad (3.15)$$

with $D_{l \rightarrow u}$ from (3.10), $E_{\max}$ from (3.12)/(3.13), $E_{\min} = -20k_B T$, and $[1 - f] \rightarrow f(E - \hbar\omega) - f(E)$ in the general case.

Remark 3.2 (The mass fork at X) *My extraction of the C&S Fig. 5 curvatures assigns $m_{u\parallel} = 0.40$, $m_{l\parallel} = 0.15$ — the opposite $\parallel$ ordering, giving $\bar{B}_X > 0$: a closed window $[\hbar\omega_{X_6^-} - \frac{B_u}{\bar{B}_X}(\hbar\omega - \mathcal{E}_g^X), E_{\max}^>]$, no geometric sub-gap tail, and a $\sqrt{\hbar\omega - \mathcal{E}_g^X}$ onset at $\mathcal{E}_g^X$ exactly. Above the gap the two branches produce the same line shape with the same singular edge (3.12) — the $\parallel$ masses enter only through $\mathcal{F}_X$, which the fit absorbs into S_X — so optical fits cannot decide the branch; the sub-gap physics and the extracted $|P_X|^2$ can. Both $\perp$ and $\parallel$ orderings of my extraction make the d band the lighter one, against the flat- d prior, so a transcription swap in my mass table is the likelier resolution; until C&S Fig. 5 is re-measured, this document follows GRW ($\bar{B}_X < 0$) and flags every place the branch matters.*

4 The L point (delegated by GRW; same machinery)

GRW cite earlier work for L ; the identical linearization gives it in three lines. The L_6^- conduction band is a minimum in both local directions ($k_{\parallel}$ along ΓL), the top d band a maximum:

$$E = \hbar\omega_u(\mathbf{k}) = \hbar\omega_{L_6^-} + \frac{\hbar^2 k_{\perp}^2}{2m_{u\perp}} + \frac{\hbar^2 k_{\parallel}^2}{2m_{u\parallel}}, \quad (4.1)$$

$$E - \hbar\omega = \hbar\omega_l(\mathbf{k}) = -\hbar\omega_{L_+} - \frac{\hbar^2 k_{\perp}^2}{2m_{l\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{l\parallel}}, \quad (4.2)$$

so that

$$\Omega_{lu} = \mathcal{E}_g^L + \bar{A}_L k_{\perp}^2 + \bar{B}_L k_{\parallel}^2, \quad \bar{B}_L = B_u + B_l > 0 \quad \text{unconditionally:} \quad (4.3)$$

the CEDS is a closed ellipsoid, existing only for $\hbar\omega \geq \mathcal{E}_g^L$ — no sub-gap tail at L , whatever the masses. The linear system is (3.5) with $-B_u \rightarrow +B_u$, $\bar{B}_X \rightarrow \bar{B}_L$, and

$$\mathcal{D}_L = A_u B_l - A_l B_u, \quad \mathcal{F}_L = \frac{\hbar^2}{2\sqrt{|\mathcal{D}_L|}} = \left[\frac{|m_{l\perp} m_{u\parallel} - m_{l\parallel} m_{u\perp}|}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2}, \quad (4.4)$$

the difference (rather than sum) structure being the fingerprint of the ellipsoidal pairing. For gold $m_{l\perp} m_{u\parallel} = 0.70 \times 0.12 = 0.084 < m_{l\parallel} m_{u\perp} = 1.03 \times 0.24 = 0.247$, so $\boxed{\mathcal{D}_L < 0}$, and the inversion (3.6) (with the sign of $\mathcal{D}$ reversing both inequalities) yields

$$k_{\parallel}^2 = \frac{\bar{A}_L}{|\mathcal{D}_L|} (E - E_{\min}), \quad D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 |\mathcal{D}_L| k_{\parallel}} = \frac{\mathcal{F}_L^2}{4\pi^2 \hbar^4 k_{\parallel}}, \quad (4.5)$$

$$E_{\min} = \hbar\omega_{L_6^-} + \frac{A_u}{\bar{A}_L} (\hbar\omega - \mathcal{E}_g^L), \quad E_{\max} = \hbar\omega_{L_6^-} + \frac{B_u}{\bar{B}_L} (\hbar\omega - \mathcal{E}_g^L), \quad (4.6)$$

with slopes $A_u/\bar{A}_L = 0.745$ and $B_u/\bar{B}_L = 0.896$ for Au. The $k_{\parallel} \rightarrow 0$ edge — the inverse-square-root singularity of (4.5) — is now the *lower* limit $E_{\min}$: exactly opposite to X , purely because $\mathcal{D}_L < 0$ while $\mathcal{D}_X > 0$. The L line shape is

$$\mathcal{I}_L(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE, \quad (4.7)$$

whose integrable divergence sits on the onset edge itself: the L edge is sharp and strong, the X edge soft and tailed — GRW’s “splitting of the interband absorption edge,” now with both halves derived from one formula set.

Remark 4.1 (L level scheme vs. the neck) *The fitted levels ($\mathcal{E}_g^L = 2.45$, $\hbar\omega_{L^+} = 1.56 \Rightarrow \hbar\omega_{L_6^-} = +0.89$ eV) put the whole window (4.6) above E_F , so the Fermi factor is inert at L and the model’s L temperature dependence enters only through broadening and gap shift. GRW’s qualitative explanation of the L singularity (onset CEDS tangent to the Fermi surface along the neck) instead needs L_6^- below E_F . Both statements cannot hold within one parabolic model; the fitted scheme is what the code implements, and the tension is recorded here deliberately.*

5 Assembly: GRW (9)–(10)

Inserting the corrected EDJDOS (3.10) into the FGR result (2.3) — note $\int d^3k \delta(\Omega - \hbar\omega)[\dots] = (2\pi)^3 \int dE D_{l \rightarrow u}(E, \hbar\omega)[\dots]$ per half-neighborhood by the definition (3.8) — the $(2\pi)^3$ cancels and

$$\begin{aligned} \epsilon_2(\hbar\omega, T) &= \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot 2 \cdot \sum_{i=X,L} N_i |P_i|^2 \mathcal{I}_i(\hbar\omega, T) \\ &= \boxed{\frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \sum_{i=X,L} N_i |P_i|^2 \mathcal{I}_i(\hbar\omega, T)} \quad [\text{GRW (9)}] \end{aligned} \quad (5.1)$$

— GRW’s Eq. (9), recreated exactly: the printed prefactor $8\pi^2 e^2 \hbar^4 / 3m^2 (\hbar\omega)^2$ emerges with no leftover constant once (i) $|P|^2$ is the squared ∇ matrix element (2.2), (ii) $D_{l \rightarrow u}$ is the corrected (3.10), (iii) spin (the explicit 2) and the half-point counts $N_X = 6$, $N_L = 8$ are kept where shown. (GRW leave N_i implicit; their fitted $|P_i|^2$ therefore means $N_i |P_i|^2$ up to normalization — immaterial for the ratio they quote.) The fitted quantities are the strengths

$$S_X = \mathcal{F}_X |P_X|^2, \quad S_L = \mathcal{F}_L |P_L|^2, \quad [\text{GRW (10)}] \quad (5.2)$$

with $|P_X/P_L|^2 = 0.370$ from their fit to Johnson–Christy. Under the corrected (4′) the combination actually determined by the data is $N_i \mathcal{F}_i^2 |P_i|^2$; it maps onto GRW’s printed S_i through the constant $2\mathcal{F}_i/\hbar^2$ of Remark 3.1, so all published numbers retain their meaning as strengths.

Parameters (levels: GRW fit; masses: C&S, with the X row subject to Remark 3.2; $\mathcal{F}$ in units of the free-electron mass):

	$\mathcal{E}_g$ (eV)	upper level (eV)	lower level (eV)	$m_{u\perp}$	$m_{u\parallel}$	$m_{l\perp}$	$m_{l\parallel}$
X (GRW branch)	1.94	$\hbar\omega_{X_6^-} = 0.17$	$\hbar\omega_{X_7^+} = 1.77$	0.19	0.15	0.31	0.40
L	2.45	$\hbar\omega_{L_6^-} = 0.89$	$\hbar\omega_{L^+} = 1.56$	0.24	0.12	0.70	1.03

	$\mathcal{D}$ (units $(\hbar^2/2m)^2$)	$\mathcal{F}/m$	window slopes	singular edge
X	+34.7	0.170	$E_{\max}^>: 0.38\text{--}0.62^\dagger$	upper ($k_\parallel \rightarrow 0$)
L	−7.86	0.357	$E_{\min}: 0.745, E_{\max}: 0.896$	lower ($k_\parallel \rightarrow 0$)

[†] 0.38 with my $\perp$ assignment ($m_{u\perp} = 0.31$), 0.62 with the fully $u \leftrightarrow l$ swapped one ($m_{u\perp} = 0.19$); $\mathcal{F}_X$ is invariant under the full swap. $\mathcal{D}_X$ and $\mathcal{F}_X$ quoted for the full-swap/GRW assignment; my assignment gives the same values by symmetry of (3.11).

Scorecard for the recreation:

GRW	here	status	defect as printed
(1)	(3.1)	recreated	—
(2)	(3.2)	recreated	—
(3)	(3.3)	recreated	—
(4)	(3.10)	corrected	prefactor $(8\pi^2\hbar^2)^{-1}\mathcal{F} \rightarrow (4\pi^2\hbar^4)^{-1}\mathcal{F}^2$; off by $2\mathcal{F}/\hbar^2$ (const.)
(5)	(3.11)	recreated	—
(6)	(3.7)	corrected	dimensionally garbled
(7)	(3.15)	recreated	occupation generalized
(8)	(3.13)	corrected	$u \parallel \leftrightarrow l \parallel$ swapped
(9)	(5.1)	recreated	N_i made explicit
(10)	(5.2)	recreated	units note (Rem. 3.1)

6 Photonic density of states and the emission spectrum

6.1 Mode counting

Two transverse polarizations, $\omega = cq$, periodic quantization:

$$\rho(\omega) = \frac{2}{(2\pi)^3} 4\pi q^2 \frac{dq}{d\omega} = \boxed{\frac{\omega^2}{\pi^2 c^3}} \quad (6.1)$$

modes per unit volume per unit angular frequency (in a host of index n , $c \rightarrow c/n$; in a structured environment $\rho \rightarrow$ the local density of states $\rho(\mathbf{r}, \omega)$ — the photonic factor of the PL factorization).

6.2 Spontaneous emission of one pair

Quantizing $\mathbf{A}$ (mode amplitudes $\sqrt{2\pi\hbar c^2/\omega V}$ in Gaussian units) and applying FGR at photon vacuum, with $\sum_\lambda \int d\Omega_q |\hat{\epsilon}_\lambda \cdot \mathbf{p}_{ul}|^2 = \frac{8\pi}{3} |\mathbf{p}_{ul}|^2$ and $|\mathbf{p}_{ul}|^2 = \hbar^2 |P|^2$:

$$W_{\text{sp}}(\omega) = \frac{4e^2\hbar\omega |P|^2}{3m^2c^3} = \underbrace{\frac{4\pi^2 e^2\hbar |P|^2}{3m^2\omega}}_{\text{matter}} \times \underbrace{\frac{\omega^2}{\pi^2 c^3}}_{\rho(\omega)}, \quad (6.2)$$

the Einstein A coefficient in the $\mathbf{A} \cdot \mathbf{p}$ form, with the photonic density of states exhibited as an exact factor. (The semiclassical field of Sec. 2 can never produce (6.2): it is the $+1$ of $\langle n+1 \rangle$, i.e. a vacuum effect — the answer to the question left open in note A1.)

6.3 The interband PL spectrum and the closure check

Summing over emitting pairs of one critical-point family (spin 2, N half-neighborhoods, EDJDOS (3.8)):

$$\boxed{R(\hbar\omega) = \rho(\omega) \frac{4\pi^2 e^2 \hbar}{3m^2\omega} 2N |P|^2 \int dE f(E, T) [1 - f(E - \hbar\omega, T)] D_{l \rightarrow u}(E, \hbar\omega)} \quad (6.3)$$

photons per unit time, volume and photon energy, with the same $D_{l \rightarrow u}$, windows and edges as in absorption — only the occupation factor differs. With the equilibrium identity $f(E)[1 - f(E - \hbar\omega)] = n_B(\hbar\omega) [f(E - \hbar\omega) - f(E)]$ (notes A2/A3), every electronic factor maps onto (5.1) and the constants cancel exactly:

$$\boxed{R(\hbar\omega) = \frac{\omega^3}{\pi^2 \hbar c^3} \epsilon_2(\omega) n_B(\hbar\omega) = \frac{\omega \rho(\omega)}{\hbar} \epsilon_2(\omega) n_B(\hbar\omega)} \quad (6.4)$$

— the van Roosbroeck–Shockley relation, and Kirchhoff’s law in the form $R_{\text{em}}/R_{\text{abs}} = e^{-\hbar\omega/k_B T}$ per mode. That (5.1) and (6.3), assembled independently, satisfy (6.4) with unit coefficient is the global consistency check of every prefactor in this document; it would fail by $2\mathcal{F}/\hbar^2$ if the printed (4) were used with any fixed matrix-element convention. Out of equilibrium ($f \rightarrow f^S, f^P$) the identity breaks, and the deviation from (6.4) is the non-equilibrium PL signal; in a nanostructure, $\rho(\omega) \rightarrow \rho(\mathbf{r}, \omega)$ in (6.3).

A Dimensional audit (Gaussian)

$[e^2] = \text{erg cm}$, $[|P|^2] = \text{cm}^{-2}$, $[\mathcal{D}] = \text{erg}^2 \text{cm}^4$, $[\mathcal{F}] = \text{g}$, $[D_{l \rightarrow u}] = \text{erg}^{-2} \text{cm}^{-3}$ (states per volume per energy²), $[\mathcal{I}] = \text{erg}^{-1} \text{cm}^{-3}$.

- (3.10): $\mathcal{F}^2/\hbar^4 k_{\parallel} \sim \text{g}^2/(\text{erg}^4 \text{s}^4 \text{cm}^{-1}) = \text{erg}^{-2} \text{cm}^{-3}$. ✓
- (5.1): $\frac{\text{erg cm} \cdot \text{erg}^4 \text{s}^4}{\text{g}^2 \text{erg}^2} \cdot \text{cm}^{-2} \cdot \text{erg}^{-1} \text{cm}^{-3} = \text{erg}^2 \text{s}^4 \text{g}^{-2} \text{cm}^{-4} = 1$. ✓
- (6.2): $\frac{\text{erg cm} \cdot \text{erg s} \cdot \text{s}^{-1} \cdot \text{cm}^{-2}}{\text{g}^2 \text{cm}^3 \text{s}^{-3}} = \text{s}^{-1}$. ✓
- (6.3): $[\rho][\cdot] \sim \text{s cm}^{-3} \cdot \text{s}^{-1} \cdot \text{erg}^{-1} \cdot \dots = \text{s}^{-1} \text{erg}^{-1} \text{cm}^{-3}$: photons/(s erg cm³). ✓
- (6.4): $[\omega\rho/\hbar] = \text{s}^{-1} \cdot \text{s cm}^{-3} \cdot \text{erg}^{-1} \text{s}^{-1} = \text{erg}^{-1} \text{cm}^{-3} \text{s}^{-1}$. ✓

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- Companion document `rosei_master.tex` — SI units, convention-general σ_c formulation, cross-repository conventions dictionary, full quantization details.